

1202SCG Calculus 1
1020ENG Engineering Mathematics 1

Week 4 Lecture

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Table of Derivatives

In the table below, n and c are constants.

Function y	Derivative $\frac{dy}{dx}$
c	0
x^n	nx^{n-1}
$\sin(x)$	$\cos(x)$
$\cos(x)$	$-\sin(x)$
e^x	e^x
$\ln(x)$	$\frac{1}{x}$

Expanded Table of Derivatives

In the table, $a > 0$ and $a \neq 1$.

Function y	Derivative $\frac{dy}{dx}$
$\tan(x)$	$\frac{1}{\cos^2(x)} = \sec^2(x)$
a^x	$a^x \ln(a)$
$\log_a(x)$	$\frac{1}{x \ln(a)}$
$\sinh(x)$	$\cosh(x)$
$\cosh(x)$	$\sinh(x)$
$\tanh(x)$	$\frac{1}{\cosh^2(x)} = \operatorname{sech}^2(x)$

Product Rule

If

$$y = uv,$$

where u and v are functions of x , then

$$y' = u'v + uv'.$$

Product Rule: Example 1

Differentiate $y = x^2 \sin(x)$.

$$u = x^2,$$

$$v = \sin(x),$$

$$u' = 2x.$$

$$v' = \cos(x).$$

$$y' = u'v + uv'$$

$$= 2x \sin(x) + x^2 \cos(x).$$

Product Rule: Example 2

Differentiate $y = (3x^2 - 1)e^x$.

$$u = 3x,$$

$$v = e^x,$$

$$u' = 3.$$

$$v' = e^x.$$

$$\begin{aligned}y' &= u'v + uv' \\&= 3e^x + 3xe^x \\&= 3e^x(1 + x).\end{aligned}$$

Product Rule: Practice

Differentiate each function.

1. $y = x^3 \cos(x)$

2. $y = (2x + 5) \ln(x)$

3. $y = x^2(4x - 1)$

4. $y = e^x \sinh(x)$

Product Rule: Answers

$$1. y' = 3x^2 \cos(x) - x^3 \sin(x)$$

$$2. y' = 2 \ln(x) + \frac{2x + 5}{x}$$

$$3. y' = 2x(4x - 1) + 4x^2 = 12x^2 - 2x$$

$$4. y' = e^x \sinh(x) + e^x \cosh(x)$$

Quotient Rule

If

$$y = \frac{u}{v},$$

where u and v are functions of x , then

$$y' = \frac{u'v - uv'}{v^2}.$$

The order in the numerator matters: $u'v - uv'$.

Quotient Rule: Example 1

Differentiate $y = \frac{\sin(x)}{x}$.

$$u = \sin(x),$$

$$v = x,$$

$$u' = \cos(x).$$

$$v' = 1.$$

$$y' = \frac{u'v - uv'}{v^2}$$

$$= \frac{x \cos(x) - \sin(x)}{x^2}.$$

Quotient Rule: Example 2

Differentiate $y = \frac{x^2 + 1}{x - 1}$.

$$u = x^2 + 1,$$

$$v = x - 1,$$

$$u' = 2x.$$

$$v' = 1.$$

$$y' = \frac{u'v - uv'}{v^2}$$

$$= \frac{2x(x - 1) - (x^2 + 1)(1)}{(x - 1)^2}$$

$$= \frac{x^2 - 2x - 1}{(x - 1)^2}.$$

Quotient Rule: Practice

Differentiate each function.

$$1. y = \frac{x^2}{x + 3}$$

$$2. y = \frac{e^x}{x^2}$$

$$3. y = \frac{\cos(x)}{2x - 1}$$

Quotient Rule: Answers

$$1. y' = \frac{2x(x+3) - x^2}{(x+3)^2} = \frac{x^2 + 6x}{(x+3)^2}$$

$$2. y' = \frac{e^x x^2 - e^x(2x)}{x^4} = \frac{e^x(x-2)}{x^3}$$

$$3. y' = \frac{-\sin(x)(2x-1) - 2\cos(x)}{(2x-1)^2}$$

Chain Rule

The chain rule is used for composite functions (when one function is inside another function).

If y is a function of u , and u is a function of x , then

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}.$$

Differentiate the outside function, then multiply by the derivative of the inside function.

Chain Rule: Powers

Differentiate $y = (3x + 1)^5$.

$$y = u^5,$$

$$u = 3x + 1,$$

$$\frac{dy}{du} = 5u^4.$$

$$\frac{du}{dx} = 3.$$

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

$$= 5(3x + 1)^4(3)$$

$$= 15(3x + 1)^4.$$

Chain Rule: Sine

Differentiate $y = 4 \sin(2x - 3)$.

$$y = 4 \sin(u),$$

$$u = 2x - 3,$$

$$\frac{dy}{du} = 4 \cos(u).$$

$$\frac{du}{dx} = 2.$$

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

$$= 4 \cos(2x - 3)(2)$$

$$= 8 \cos(2x - 3).$$

Chain Rule: Cosine

Differentiate $y = \cos(x^2)$.

$$y = \cos(u),$$

$$u = x^2,$$

$$\frac{dy}{du} = -\sin(u).$$

$$\frac{du}{dx} = 2x.$$

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

$$= -\sin(x^2)(2x)$$

$$= -2x \sin(x^2).$$

Chain Rule: Exponential Functions

Differentiate $y = e^{x^2+1}$.

$$y = e^u,$$

$$u = x^2 + 1,$$

$$\frac{dy}{du} = e^u.$$

$$\frac{du}{dx} = 2x.$$

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

$$= e^{x^2+1}(2x)$$

$$= 2xe^{x^2+1}.$$

Chain Rule: Logarithmic Functions

Differentiate $y = \ln(4x - 1)$.

$$y = \ln(u),$$

$$u = 4x - 1,$$

$$\frac{dy}{du} = \frac{1}{u}.$$

$$\frac{du}{dx} = 4.$$

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

$$= \frac{1}{4x - 1} (4)$$

$$= \frac{4}{4x - 1}.$$

Chain Rule: Practice

Differentiate each function.

1. $y = (5x - 2)^4$

2. $y = \sqrt{x^2 + 3}$

3. $y = 3 \cos(4x)$

4. $y = e^{2x^3}$

5. $y = \ln(x^2 + 1)$

Chain Rule: Answers

$$1. \frac{dy}{dx} = 20(5x - 2)^3$$

$$2. \frac{dy}{dx} = \frac{x}{\sqrt{x^2 + 3}}$$

$$3. \frac{dy}{dx} = -12 \sin(4x)$$

$$4. \frac{dy}{dx} = 6x^2 e^{2x^3}$$

$$5. \frac{dy}{dx} = \frac{2x}{x^2 + 1}$$

Combining Product and Chain Rules

Differentiate $y = x^2 \sin(3x)$.

$$u = x^2,$$

$$v = \sin(3x),$$

$$u' = 2x.$$

$$v' = 3 \cos(3x).$$

$$y' = u'v + uv'$$

$$= 2x \sin(3x) + 3x^2 \cos(3x).$$

Combining Quotient and Chain Rules

Differentiate $y = \frac{e^{2x}}{x+1}$.

$$u = e^{2x},$$

$$v = x + 1,$$

$$u' = 2e^{2x}.$$

$$v' = 1.$$

$$y' = \frac{u'v - uv'}{v^2}$$

$$= \frac{2e^{2x}(x+1) - e^{2x}}{(x+1)^2}$$

$$= \frac{e^{2x}(2x+1)}{(x+1)^2}.$$

Implicit Differentiation

Some equations describe a relationship between x and y without explicitly writing y as a function of x .

For example,

$$x^2 + y^2 = 25.$$

Simply differentiate both sides with respect to x :

$$\frac{d}{dx} (x^2 + y^2) = \frac{d}{dx}(25).$$

Differentiating Terms Containing y

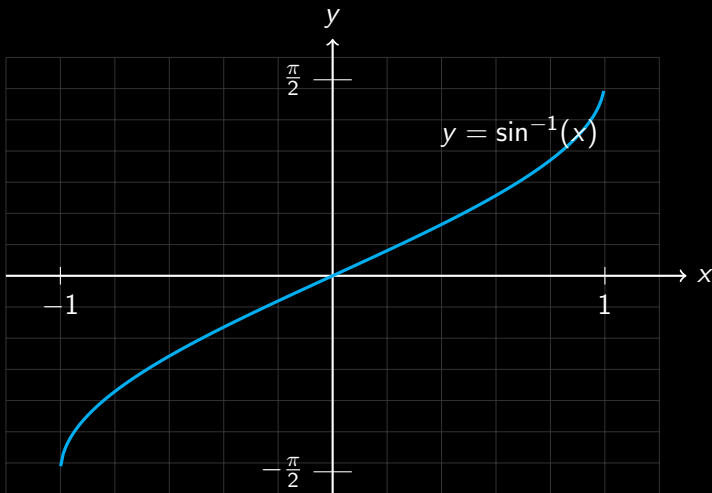
Use the chain rule whenever a function of y is differentiated.

$$\frac{d}{dx}(y) = \frac{dy}{dx},$$

$$\frac{d}{dx}(y^2) = \frac{d}{dy}(y^2) \frac{dy}{dx} = 2y \frac{dy}{dx},$$

$$\frac{d}{dx}(\sin y) = \frac{d}{dy}(\sin y) \frac{dy}{dx} = \cos(y) \frac{dy}{dx}.$$

The Inverse Sine Function



The function is increasing, so its derivative should be positive.

Derivative of Inverse Sine

What is the derivative of $\sin^{-1}(x)$ (with respect to x).

$$\text{Let } y = \sin^{-1}(x)$$

$$\sin(y) = x.$$

Differentiate both sides implicitly:

$$\frac{d}{dx}(\sin(y)) = \frac{d}{dx}(x)$$

$$\cos(y) \frac{dy}{dx} = 1,$$

$$\frac{dy}{dx} = \frac{1}{\cos(y)}.$$

Derivative of Inverse Sine

Use the identity

$$\sin^2(y) + \cos^2(y) = 1$$

$$\cos^2(y) = 1 - \sin^2(y)$$

$$\cos(y) = \sqrt{1 - \sin^2(y)}$$

$$\cos(y) = \sqrt{1 - x^2}.$$

$$\therefore \frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}}, \quad -1 < x < 1.$$

Note that we ignored the negative case as we saw from the graph that the gradient of $\sin^{-1}(x)$ was always positive.

Derivative of the Natural Logarithm

Let

$$y = \ln(x).$$

Equivalently,

$$e^y = x.$$

Differentiate both sides implicitly:

$$e^y \frac{dy}{dx} = 1,$$

$$\frac{dy}{dx} = \frac{1}{e^y}$$

$$= \frac{1}{x}.$$

This derivative is defined for $x > 0$.

Implicit Differentiation: A Circle

Find $\frac{dy}{dx}$ if

$$x^2 + y^2 = 25.$$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(25)$$

$$2x + 2y \frac{dy}{dx} = 0$$

$$2y \frac{dy}{dx} = -2x$$

$$\frac{dy}{dx} = -\frac{x}{y}.$$

Tangent to an Implicit Curve

Find the tangent to $x^2 + y^2 = 25$ at $(3, 4)$.

$$\frac{dy}{dx} = -\frac{x}{y},$$

$$m = \left. \frac{dy}{dx} \right|_{(3,4)} = -\frac{3}{4}.$$

Using $y - y_1 = m(x - x_1)$,

$$y - 4 = -\frac{3}{4}(x - 3)$$

$$y = -\frac{3}{4}x + \frac{25}{4}.$$

Implicit Differentiation

Find $\frac{dy}{dx}$ if

$$xy + y^2 = 6.$$

The term xy requires the product rule:

$$\frac{d}{dx}(xy) + \frac{d}{dx}(y^2) = 0$$

$$y + x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$$

$$\frac{dy}{dx}(x + 2y) = -y$$

$$\frac{dy}{dx} = -\frac{y}{x + 2y}.$$

Implicit Differentiation: Practice

Find $\frac{dy}{dx}$ for each equation.

1. $x^2 + 3y^2 = 12$

2. $x^3 + y^3 = 9$

3. $x^2 + xy + y^2 = 7$

Implicit Differentiation: Answers

$$1. \frac{dy}{dx} = -\frac{x}{3y}$$

$$2. \frac{dy}{dx} = -\frac{x^2}{y^2}$$

$$3. \frac{dy}{dx} = -\frac{2x + y}{x + 2y}$$

Choosing a Differentiation Rule

Before differentiating, identify the structure.

- ▶ Multiplication of functions $\longrightarrow$ product rule
- ▶ Division of functions $\longrightarrow$ quotient rule
- ▶ A function inside another $\longrightarrow$ chain rule
- ▶ An equation involving both x and y $\longrightarrow$ implicit differentiation

Simplify the final derivative where appropriate.

An Undefined Gradient

The curve is defined implicitly by

$$x^2y + y^3 = x.$$

1. Find an expression for $\frac{dy}{dx}$.
2. Evaluate the gradient at $(0, 0)$.
3. Determine the equation of the tangent line at $(0, 0)$.
4. Determine the equation of the normal line at $(0, 0)$.

An Undefined Gradient: Solution

Differentiate both sides with respect to x :

$$2xy + x^2 \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 1,$$

$$\frac{dy}{dx} (x^2 + 3y^2) = 1 - 2xy,$$

$$\frac{dy}{dx} = \frac{1 - 2xy}{x^2 + 3y^2}.$$

At $(0,0)$, the numerator is 1 and the denominator is 0, so the gradient is undefined and the tangent is vertical.

Tangent: $x = 0$, Normal: $y = 0$.

The Curve Near the Origin

